ASTR 457: Foundations of Data Science in Astronomy

Fall 2026

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Lecture: Astronomy 134, Tue & Thur, 1230 – 1350
: https://github.com/gnarayan/ast457_2026_Fall
Office Hours: Astronomy 129, by appointment

COURSE DESCRIPTION & LEARNING GOALS

This 16 week course (3 contact hours) will cover a number of statistical techniques that are relevant to astrophysical studies. These include robust statistics, regression, model building and hypotheses testing, MCMC methods, parameter estimation, time series analysis, clustering and dimensionality reduction, hierarchical modeling, and modern machine learning, including neural networks, foundation models, and the large language models you are already using. We will also cover best practices for writing code and version control. These techniques are ubiquitous in science and industry. My goal is to provide a survey of these techniques, together with realistic problems, so that you see how they work and what their implicit assumptions are.

This course is built for the agentic AI era. AI assistants can now produce a complete, plausible-looking analysis of most well-specified problems in seconds. AI has inverted which skills matter. The scarce skill is no longer producing a fit; it is knowing whether the fit is *right*: whether the assumptions hold, whether the uncertainties are calibrated, and whether the conclusions survive scrutiny. Accordingly, you will use AI tools openly and document how you used them, and you will be graded on the judgment, verification, and understanding that you yourself bring to the problem. The TA is **Abha Vishwakarma** (office hours Fri. 1:00–2:00 pm or by appointment, over Zoom; Wed. 11:00–noon until Oct. 2).

By the end of this semester you should be able to:

- Apply a variety of existing models in astronomy, and interpret the results and compare to existing literature.
- Identify when existing models are inadequate descriptions of the astrophysical circumstances under study, and develop new models using your knowledge of statistical and computational techniques.
- Report inferences from your models with appropriate uncertainties, visualizations that convey the robustness of the model you selected, and describe what you did in a manner appropriate for high-impact scientific journals.
- Critically evaluate an analysis produced by someone else (a collaborator, a published paper, or an AI system), identify its errors and unstated assumptions, and verify or refute its conclusions with quantitative evidence.

To achieve these outcomes, students enrolled for 3 credit hours are expected to work 6 hours/week outside instruction time.

PREREQUISITES

Required: ASTR 210 & 310, to provide the necessary background in modern astronomy and Python programming. These have additional calculus & physics prerequisites.

Recommended: Prior coursework in undergraduate statistics and undergraduate linear algebra (e.g., MATH 227 or 257). You will also need a computer with a working `conda` and `git` installation for much of the coursework.

TEXTS & READINGS

- "Statistics, Data Mining, and Machine Learning in Astronomy", Ž. Ivezić, A. Connolly, J. T. VanderPlas & A. Gray (Princeton University Press, ISBN 9780691198309)
- "Python Data Science Handbook", J T. VanderPlas (O'Reilly Media)

Copies of both books are available online (the previous edition for ICVG).

ICVG: Through O'Reilly (free registration with your illinois.edu email required)
`https://learning.oreilly.com/library/view/statistics-data-mining/9780691151687/`
or through JSTOR: `https://www.jstor.org/stable/j.ctt4cgbdj`

VdP: On GitHub `https://jakevdp.github.io/PythonDataScienceHandbook/`

Other Resources:

"Modern Statistical Methods for Astronomy", E. Feigelson with J. Babu (**FB**) is detailed, though focuses on using R, and is available for free as an e-book through NASA ADS. You may also find "Bayesian Models for Astrophysical Data", J. M. Hilbe, R. S. de Souza, & E. E. O. Ishida helpful (1st edition, Cambridge University Press, ISBN 9781107133082). I recommend "Data Analysis: A Bayesian Tutorial", D. S. Sivia and J. Skilling (2nd edition, Oxford University Press) if you need a quick refresher on prerequisite material, including the linear algebra we will lean on for Gaussian processes and dimensionality reduction.

WORKING WITH AI: THE GROUND RULES

Using AI assistants (Copilot, Claude, ChatGPT, Gemini, or whatever ships next month) is **allowed and expected** on all labs and take-home exams. Professional research in 2026 involves these tools, and "realistic research conditions," the founding ethos of this course, now includes them. The University provides every student free access to Microsoft Copilot, so every assignment is designed to be completable with free-tier tools; no one needs to pay for a subscription to do well in this course.

The rules:

- **Document it.** Every lab and exam submission includes a short AI-use appendix: which tools you used, what you asked them for, what they got wrong, and how you found out. "I did not use AI for this assignment" is a perfectly acceptable appendix, if it's true.
- **You are the author.** You are graded on verification and judgment: checking the assumptions and catching the errors. Turning in unverified AI output is turning in someone else's work with your name on it. As you'll discover in this course, unverified AI output is frequently, confidently wrong.

- **You must be able to defend it.** Any submission can be selected for a short oral defense: you and your notebook, a few minutes of questions about why you made the choices you made. If you cannot explain your own submission, it isn't your submission, and it will be graded accordingly.
- **The line.** Discussing problems with classmates: encouraged. Using AI and documenting it: encouraged. Sharing solutions with each other: cheating. Misrepresenting AI work as verified, or fabricating your AI-use appendix: cheating, of the referral-to-the-Senate-Committee variety.

GRADING

Your grade combines two tracks. **Track A** (quizzes, oral components) is done live and AI-free, and certifies the statistical reasoning you carry in your own head. **Track B** (labs, take-home exam components, the capstone) is open-book, open-AI, and documented, and certifies that you can do, and stand behind, a real analysis. You need both to do well; neither can be delegated to a machine. Attendance is at your own discretion. Extra credit is rare; the one planned opportunity is the Sep. 8 Astronomy Colloquium (see Important Dates). This course awards 3 credit hours; graduate students may register for either 3 or 4 credit hours. At 3 credit hours, graduate students complete exactly the same assessments as undergraduates; the 4 credit hour option adds an expanded capstone (see below). **All coursework uses Github, not a LM like Canvas, to replicate realistic research conditions.** Periodic overall grade updates are sent via email. You are welcome to discuss your grades and your work in the course with me during office hours.

Points:

• Labs (roughly weekly, AI-in-the-loop, documented):	25%
• In-class quizzes (short, written, AI-free):	15%
• Midterm (take-home, graded like a lab):	20%
• Final (take-home, graded like a lab):	20%
• Capstone (notebook, video & lightning talk):	20%

Labs (roughly ten, equally weighted). Each lab hands you a dataset generated specifically for you: your data, your noise, your truth. Your grade depends on recovering parameters you cannot look up and on whether your reported uncertainties actually cover the truth at the rate they claim. Several labs include a supplied AI-generated analysis that is subtly (or not so subtly) wrong; your job is the referee report: find the flaws, demonstrate them quantitatively, and fix them. You may drop ONE lab from your total, for whatever reason, no questions asked (and if you don't elect to, I'll drop your lowest). Your lowest quiz is likewise dropped.

The capstone is a semester project applying a technique from this course (or beyond it) to a real astronomical dataset. You deliver three things: a worked Jupyter notebook; a recorded 10-minute presentation uploaded to YouTube (an unlisted link is fine; turn captions on) and submitted with your notebook by Noon on Mon Dec 7; and a 4-minute in-class lightning talk in the final week (ten talks per session, Dec 3 and Dec 8). The video carries the depth; the lightning talk is your elevator pitch, and questions on your capstone are fair game in the lightning-talk Q&A. Topics are chosen with me by mid-October. Graduate students taking the 4 credit hour option additionally deliver a written survey of how their technique is used in the current astrophysics and statistics literature, record a 15-minute video (rather than 10) covering that survey, and are graded to a correspondingly higher standard; this expanded capstone is the additional work that earns the fourth credit hour.

This course (like every other one at UIUC...) uses a plus (+) and minus (−) grading scale for course grades. Grades are rounded up (i.e. `ceil()`) to the nearest integer.

97-100=A+; 93-96=A; 90-92=A-; 87-89=B+; 83-86=B; 80-82=B-; 77-79=C+; 73-76=C; 70-72=C-; 67-69=D+; 63-66=D; 60-62=D-; 0-59=F

COURSE POLICIES

I've outlined standards for this course below. Times listed in this syllabus are US/Central throughout. If something is not covered by my policies, please discuss it with me. My contact information is at the beginning of this syllabus.

Assignment & Exam Policies: Labs and the take-home midterm and final are open book, open AI (documented, per the ground rules above). You may work in groups, and may discuss the assignments and ways to tackle them, but you must write/code your solution independently and run your analysis on your own dataset. Your data are unique to you, so your numbers and plots will not match anyone else's. If they do, we will be having a very short conversation.

Labs/exams will be posted to the course GitHub repo on Thursdays. Make a fork of the repo, create a folder named for your NetID inside `submissions/` for your work, write/code up your solution as directed in the assignment, commit, and open a pull request when you are satisfied with your work before Noon the following Wednesday.

Quizzes (roughly six, equally weighted) are brief written checks (10–15 minutes) at the start of class, closed-everything, roughly every other week, covering the reasoning behind what we've done: reading a corner plot, or saying why an estimator misbehaves. No computation, no AI, no notes: these certify what you carry in your own head.

Oral defenses are short (5–10 minute) scheduled conversations with me or the TA about work you submitted. Each of you defends one lab or the midterm during the semester, drawn at random; capstone questions come in the lightning-talk Q&A. The questions are about your choices and your checks. They are a rehearsal for every seminar, group meeting, and thesis defense you will ever attend.

The midterm and final will be posted online, and will be due by Noon six days later (for the final, Dec 15, during finals week). If you have a conflict with the exam dates, please contact me as soon as possible. Make up examinations will have different questions. Exams include all material covered prior, and will require a more substantial time commitment than the weekly labs. While I am open to accommodating students who need to take these exams at different times for any reason, all grades for the course are due to the Registrar by Dec. 22, 2026, and I cannot provide extensions beyond that date, unless there are absolutely extenuating circumstances.

Grades of Incomplete: Incomplete (I) grades follow College of LAS policies and procedures: the authority to authorize an incomplete rests with the college, not with me. Incompletes are appropriate only where unexpected emergencies prevent you from completing the course and the remaining work can be completed the next semester, and documentation must be provided. Incomplete work must be finished by the 10th day of instruction in the Spring 2027 semester, else the "I" will automatically be recorded as a "F" on your transcript.

Late or Missed Assignments: Labs are assigned on Thursday and due the following Wednesday by Noon. If you know that you will be turning an assignment in late please notify me in advance. A full letter grade will be deducted for each day an assignment is late until a "F" grade is achieved, unless you have a documented medical excuse or you have notified me of other extenuating circumstances. Remember that you may drop ONE lab and one quiz from your total, for whatever reason, no questions asked. Missed quizzes cannot be made up (that's what the drop is for); missed oral defenses can be rescheduled once with advance notice. Per the university's Student Code, I will reasonably accommodate class absences; if I ask for an absence letter, you can obtain one from the Connie Frank CARE Center.

Accessibility Accommodation: It is my goal that this class be an accessible and welcoming experience for all students, including those with disabilities that may impact learning in this class. If the design of this course poses barriers to you effectively participating and/or demonstrating learning in this course, please meet with me, with or without an Accessibility Services accommodation letter, to discuss reasonable options or adjustments. You are welcome to talk to me at any point in the semester about course design concerns, but it is always best if we can talk at least one week prior to the need for any modifications.

During our discussion, I may suggest the possibility/necessity of your contacting the Office of Disability Resources and Educational Services (1207 S. Oak St., Champaign, IL 61820; 217-333-1970) disability@illinois.edu; `http://disability.illinois.edu/`) to talk about academic accommodations.

Academic Integrity: Plagiarism: **Don't.** Copying a classmate's solution is cheating. Using AI is allowed; lying about it is not. Fabricating your AI-use appendix, presenting unverified machine output as your own verified analysis, or having someone (or something) else sit your quiz or defense are all academic integrity violations, and will result at least in an "F" for that work, possibly an "F" for the course, and referral to the Senate Committee for Student Discipline. Read the University of Illinois' policy on plagiarism.

I am confident in each of your ability to tackle the course work. The assessment in this course is designed so that honest work is also the path of least resistance: your dataset is yours alone, and documented AI use costs you nothing. If you feel you need help with material, come see me during office hours.

Classroom Behavior: I expect you to live up to your roles as student-scholars. Students must follow the University of Illinois' standards for personal and academic conduct. Proper conduct entails creating a **positive** learning experience for all students, regardless of sex, race, religion, sexual orientation, social class, or any other feature of personal identification; therefore, **sexist, racist, prejudicial, homophobic, or other derogatory remarks will not be tolerated.**

Syllabus Amendment: This syllabus may be amended or modified in any way upon notice, with the version on GitHub being authoritative. Most such changes will affect the tentative schedule, but be sure that you know if any due dates change. An accessible, screen-reader-friendly version of this syllabus is maintained as 457_FDS_Syllabus.md in the course repo and is updated whenever this PDF is.

Reaching Me: The fastest ways to get help are office hours and any time my door is open. Email works too: I respond promptly on weekdays, but I will not respond to email over the weekend (6:00 pm Friday to 8:00 am Monday), though I will take into account when you sent it. On the weeks I am traveling, expect Zoom office hours and slower replies. This class should be fun, and the primary learning outcome is that you grow as a scientist; if something is getting in the way of that, tell me early.

Important Dates:

- Aug. 25, 2026: First day of class
- Sep. 8, 2026: Astronomy Colloquium by Kareem El-Badry, who led the Gaia BH1 black-hole discovery you'll meet on day one. **Extra credit, +5 points on Quiz 1:** attend (there will be a signup sheet) or watch the YouTube recording afterward, and turn in at least a page of notes including two questions you had, by Noon Tue Sep. 15
- Oct. 8, 2026: Midterm take-home posted (due Oct. 14 by Noon)
- Oct. 20, 2026: Capstone topics due (chosen with me)
- Nov. 24 & 26, 2026: Fall break, no class
- Dec. 3 & 8, 2026: Capstone lightning talks in class
- Dec. 7, 2026: Capstone notebooks & videos due by Noon
- Dec. 9, 2026: Last day of instruction
- Dec. 9, 2026: Final take-home posted (due Dec. 15 by Noon)

CLASS SCHEDULE FALL 2026 (subject to revision)

Instructor travel: I am at conferences or giving invited talks during the weeks of Sep 1, Sep 8, Sep 29, Oct 13, Nov 3, and Nov 17 — shaded grey below. The affected sessions will be led by a substitute or held on Zoom; watch your email for the arrangements each week.

- **Aug 25, 27**
  First steps, crash course in python; how this course treats AI, including a live demonstration of an AI agent solving (and botching) a problem set
- **Sep 1, 3**
  Probability distributions, descriptive statistics, the Central Limit theorem and when it doesn't hold, robust statistics, and hypothesis testing (ICVG Ch. 3, FB Ch. 2)
- **Sep 8, 10** *(the TA proctors Quiz 1 on Tue)*
  Statistical inference, frequentist properties such as unbiasedness & the Cramér–Rao bound, consistency, asymptotic limits, mean-squared errors (ICVG Ch. 4, FB Ch. 3)
- **Sep 15, 17**
  Maximum likelihood estimation and applications, ranting about minimizing χ^2 (ICVG Ch. 4)
- **Sep 22, 24**
  Regression & Inference: ordinary least squares, generalized least squares, orthogonal distance regression vs generative modeling of data (ICVG Ch. 8, FB Ch. 7)
- **Sep 29, Oct 1** *(GN via Zoom Sep 29; guest lecture Padma Venkatraman Oct 1)*
  Bayes in practice, sampling and Markov Chain Monte Carlo methods (ICVG Ch. 5). **Regression review test in class Sep 29** (your own notebook, talking allowed; counts as one lab; pull request due Wed Sep 30 by Noon).

- **Oct 6, 8**
  Building models, model comparison and the Bayes factor, effective sampling techniques, estimating parameters & uncertainties, posterior predictive checks, other MCMC wizardry (ICVG Ch. 8). **Midterm posted Oct 8.**
- **Oct 13, 15**
  Visualization as verification: plots that catch broken models before referees do (VdP Ch. 4). **Midterm due Oct 14 by Noon.**
- **Oct 20, 22**
  Visualization to communicate (Oct 20, VdP Ch. 4), then time-series analysis (ICVG Ch. 10, FB Ch. 11) and Gaussian processes (ICVG Ch. 8.10, readings from Rasmussen & Williams) from Oct 22. **Quiz 3 opens class on Thu Oct 22** (Bayes and MCMC, Days 12–14).
- **Oct 27, 29**
  Probabilistic Graphical Models (PGMs) & hierarchical Bayes (Readings from Hilbe, de Souza & Ishida)
- **Nov 3, 5** *(on Zoom; the TA proctors Quiz 4 on Tue)*
  The ABCs of not having a likelihood function: ABC and simulation-based inference (Readings from Hilbe, de Souza & Ishida)
- **Nov 10, 12**
  Machine learning: tree methods, supervised classification, outliers, imbalanced & missing data; unsupervised clustering, density estimation & dimensionality reduction (ICVG Ch. 6, 7, 9, VdP Ch. 5)
- **Nov 17, 19** *(traveling Tue only)*
  Neural networks to foundation models: how modern astronomical surveys use deep learning, with case studies from time-domain astronomy
- **Nov 24, 26**
  Fall break (Nov 21–29), no class. Nothing is due over break: the lab posted Nov 19 is due Wed Dec 2.
- **Dec 1, 3**
  Large language models and AI agents: how they work, how they fail, and verification as a statistical problem, applying everything this course taught you to the machine itself (Tue and the first half of Thu). **First ten lightning talks Thu Dec 3; capstone notebooks & videos due Mon Dec 7 by Noon.**
- **Dec 8**
  Remaining ten lightning talks; course wrap. **Final posted Dec 9.**
- **Dec. 15th**
  Final take-home due by Noon (finals period; grades are due to the Registrar Dec 22).